ERDŐS PROBLEM 282 (GRAHAM'S VARIANT): NON-TERMINATION OF THE GREEDY ALGORITHM ON $1/119 \pmod 3$

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1. The problem

Fix an arithmetic progression $A = \{n \geq 1 : n \equiv a \pmod d\}$ with $d > 1$. Given a rational $x \in (0, 1)$, the restricted greedy algorithm repeatedly chooses the smallest unused integer $n \in A$ with $n \geq 1/x$ and replaces x by $x - 1/n$, terminating if and only if the remainder reaches 0.

Graham [1] proved that $m/n \in \mathbb{Q}_{>0}$ is a finite sum of distinct unit fractions with denominators in A if and only if

$$\gcd\left(\frac{n}{\gcd(n, a, d)}, \frac{d}{\gcd(a, d)}\right) = 1.$$

In [2, p. 31] (see [3, 4]), Erdős and Graham asked whether the restricted greedy algorithm always terminates whenever Graham's condition holds. The answer is no.

Theorem. *Let $A = \{n \geq 1 : n \equiv 1 \pmod 3\}$ ($a = 1$, $d = 3$) and $x = 1/119$. Although $\gcd(119, 3) = 1$ (and $1/119 = 1/136 + 1/952$ with $136, 952 \equiv 1 \pmod 3$), the greedy algorithm starting from $1/119$ with denominators in A never terminates.*

2. Why greedy gets trapped: the +1 numerator loop

When all positive integers are allowed ($A = \mathbb{N}$), Fibonacci observed in 1202 that picking $n = \lceil 1/x \rceil$ strictly decreases the numerator of x at every step, forcing termination. When $A = \{n \equiv 1 \pmod 3\}$, the opposite happens whenever the current remainder has denominator 1 less than a multiple of 3 × (numerator):

- Starting at $1/119$: both $119 \equiv 2$ and $120 \equiv 0 \pmod 3$ are forbidden, so greedy must pick 121, leaving
$$\frac{1}{119} - \frac{1}{121} = \frac{2}{14{,}399}.$$
- At $2/14{,}399$: dividing 14,399 by 2 gives 7,199.5, whose ceiling 7,200 is a forbidden multiple of 3. Greedy must overshoot by +1 to 7,201, leaving $3/103{,}687{,}199$.
- Each +1 overshoot raises the numerator by +1: $1 \to 2 \to 3 \to 4 \to 5 \to 6$. At step 6, the unreduced numerator 7 cancels against a factor of 7 in the 6th picked denominator, resetting the remainder back to a unit fraction $1/(6c_{\text{new}} - 1)$ whose parameter c_{new} satisfies the exact same modular rules as $c = 20$.

3. Exact six-step algebraic derivation in c alone

Let $c \geq 1$ be an integer satisfying

$$c \equiv 6 \pmod 7 \quad \text{and} \quad \big(c \equiv 0 \pmod 5 \ \text{ or } \ c \equiv 18 \pmod{25}\big), \tag{1}$$

and suppose that all integers $> 6c$ congruent to 1 (mod 3) are still unused (initially $c = 20$, giving $1/(6c - 1) = 1/119$, with every $n \equiv 1 \pmod 3$ available). At each greedy step we divide the current denominator by the current numerator, round up, and advance to the smallest integer $\equiv 1 \pmod 3$ (noting that each newly picked denominator strictly exceeds all previously removed ones, so no collision with used denominators ever occurs):

Step 1. From $\dfrac{1}{6c - 1}$, dividing $6c - 1$ by 1 gives $6c - 1 \equiv 2 \pmod 3$, so greedy picks $6c + 1$:

$$\frac{1}{6c - 1} - \frac{1}{6c + 1} = \frac{2}{(6c - 1)(6c + 1)} = \frac{2}{36c^2 - 1}.$$

Step 2. From $\dfrac{2}{36c^2-1}$, dividing by 2 gives $18c^2-\frac{1}{2}$, whose ceiling $18c^2$ is $\equiv 0 \pmod 3$. Greedy picks $18c^2+1$:

$$\frac{2}{36c^2-1}-\frac{1}{18c^2+1}=\frac{3}{(36c^2-1)(18c^2+1)}=\frac{3}{648c^4+18c^2-1}.$$

Step 3. Dividing $648c^4+18c^2-1$ by 3 gives $216c^4+6c^2-\frac{1}{3}$, whose ceiling is $\equiv 0 \pmod 3$. Greedy picks $216c^4+6c^2+1$:

$$\frac{3}{648c^4+18c^2-1}-\frac{1}{216c^4+6c^2+1}=\frac{4}{139{,}968c^8+7{,}776c^6+540c^4+12c^2-1}.$$

Step 4. Dividing by 4 gives $34{,}992c^8+1{,}944c^6+135c^4+3c^2-\frac{1}{4}$; every coefficient is a multiple of 3, so greedy picks $34{,}992c^8+1{,}944c^6+135c^4+3c^2+1$, leaving

$$\frac{5}{9c^2(36c^2+1)(324c^4+9c^2+1)(46{,}656c^8+2{,}592c^6+180c^4+4c^2+1)-1}.$$

Step 5. Dividing that denominator by 5 gives

$$\frac{9}{5}\,c^2(36c^2+1)(324c^4+9c^2+1)(46{,}656c^8+2{,}592c^6+180c^4+4c^2+1)-\frac{1}{5}.$$

By (1), either $5 \mid c$ (so $5 \mid c^2$) or $c \equiv 18 \equiv -7 \pmod{25}$ (so $c^2 \equiv -1 \pmod{25}$ and $36c^2+1 \equiv -35 \pmod{25}$ is divisible by 5). Thus the 5 cancels cleanly and the factor 9 keeps the ceiling $\equiv 0 \pmod 3$. Greedy adds $+1$, picking

$$\frac{1}{5}\big(4{,}897{,}760{,}256c^{16}+544{,}195{,}584c^{14}+52{,}907{,}904c^{12}+2{,}939{,}328c^{10} \\ +224{,}532c^8+9{,}072c^6+441c^4+9c^2+5\big)$$

and leaving numerator 6 over

$$\frac{9}{5}\,c^2(36c^2+1)(324c^4+9c^2+1)(46{,}656c^8+\cdots+1)(4{,}897{,}760{,}256c^{16}+\cdots+9c^2+4)-1.$$

Step 6. Dividing that denominator by 6 gives

$$\frac{3}{10}\,c^2(36c^2+1)(324c^4+9c^2+1)(46{,}656c^8+\cdots+1)(4{,}897{,}760{,}256c^{16}+\cdots+441c^4+9c^2+4)-\frac{1}{6}.$$

Here 4 always divides $c^2(324c^4+9c^2+1)(4{,}897{,}760{,}256c^{16}+\cdots+441c^4+9c^2+4)$: if c is even then $4 \mid c^2$, while if c is odd both parenthesized factors are even ($324+9+1=334$ and $441+9+4=454$). Hence the 2 in 10 cancels with a factor of 2 to spare, so the ceiling is an even multiple of 3. Greedy adds $+1$, picking

$$\frac{7{,}996{,}018{,}508{,}417{,}728{,}512c^{32}+1{,}776{,}893{,}001{,}870{,}606{,}336c^{30}+\cdots+615c^4+12c^2+10}{10}.$$

Subtracting its reciprocal produces unreduced numerator 7 over $(6c-1)$ times all six picked denominators. Because $c \equiv 6 \pmod 7$, we have $6c-1 \equiv 0 \pmod 7$, so every step's running denominator is $0 \bmod 7$, making this 6th picked denominator congruent to $(0+1)/6+1 \equiv 6+1 \equiv 0 \pmod 7$ as well. Cancelling 7 resets the numerator to 1, transforming $\dfrac{1}{6c-1}$ in 6 greedy steps into $1/D(c)$, where

$$\begin{aligned} D(c) &= \frac{1}{350}(6c-1)(6c+1)(18c^2+1)(216c^4+6c^2+1) \\ &\quad\times(34{,}992c^8+1{,}944c^6+135c^4+3c^2+1)(4{,}897{,}760{,}256c^{16}+\cdots+9c^2+5) \\ &\quad\times(7{,}996{,}018{,}508{,}417{,}728{,}512c^{32}+\cdots+12c^2+10) \\ &= 6\cdot\frac{63{,}936{,}311{,}986{,}958{,}875{,}890{,}717{,}097{,}301{,}737{,}734{,}144\,c^{64}+\cdots+100c^2+100}{700}-1. \end{aligned}$$

4. Why the invariants repeat forever

Read the new parameter c_{new} directly off the resulting expression:

(1) *Integrality and* $\equiv 5 \pmod 6$ *shape:* In the product, $6c-1 \equiv 5 \pmod 6$, all six picked denominators are even multiples of 3 plus 1 ($\equiv 1 \pmod 6$), and $7 \equiv 1 \pmod 6$. Thus the reduced denominator is an integer $\equiv 5 \pmod 6$, i.e., $6c_{\text{new}}-1$ for an integer $c_{\text{new}} > c$ (with all $> 6c_{\text{new}}$ elements of A still untouched).

(2) *Preservation of* $c_{\text{new}} \equiv 6 \pmod 7$*:* Both $6c-1$ and the 6th picked denominator are multiples of 7; cancelling the numerator 7 removes only one of the two factors of 7, so $6c_{\text{new}} - 1 \equiv 0 \pmod 7$, i.e., $c_{\text{new}} \equiv 6 \pmod 7$.

(3) *Alternation* $0 \bmod 5 \leftrightarrow 18 \bmod 25$*:*
- If $5 \mid c$ (as at $c = 20$), then $25 \mid c^2$ and $625 \mid c^4, c^6, \ldots, c^{64}$, so in $c_{\text{new}} = (\cdots + 5{,}269c^4 + 100c^2 + 100)/700$ every c-term vanishes modulo 25 after dividing by $700 = 25 \times 28$, leaving $c_{\text{new}} \equiv 100/700 = 1/7 \equiv 18 \pmod{25}$.
- If $c \equiv 18 \pmod{25}$, substituting $c^2 \equiv -1 \pmod{25}$ makes the degree-64 numerator divisible by 125, so dividing by 700 gives $c_{\text{new}} \equiv 0 \pmod 5$.

By induction on the number of 6-step rounds $K \geq 0$, starting at $c = 20$ ($x = 1/119$) produces a strictly positive remainder $1/(6c_K - 1) > 0$ after every $6K$ steps, so the greedy algorithm never reaches 0. This proves the Theorem. □

Remark. The argument above is formalized in Lean 4 in `FormalConjectures/ErdosProblems/282.lean` (theorem `Erdos282.erdos_282.variants.graham`).

References

[1] R. L. Graham, *On finite sums of unit fractions*, Proc. London Math. Soc. (3) **14** (1964), 193–207.

[2] P. Erdős and R. L. Graham, *Old and new problems and results in combinatorial number theory*, Monographies de L'Enseignement Mathématique, vol. 28, Université de Genève, 1980.

[3] *Erdős Problem #282*, `https://www.erdosproblems.com/282`.

[4] The Formal Conjectures Authors, *FormalConjectures/ErdosProblems/282.lean*, `https://github.com/google-deepmind/formal-conjectures`.